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The `mousik` package*

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Abstract

This is a music-related package focused on writing music fragments. It works specially well with small examples, such as excerpts for theoretical articles. It is easier than other music writing methods in L^AT_EX because it is based on the `tikz` package.

Keywords

mousik, music, music theory, music notation, music writing, score, sheet, tikz diagram, bar, clef, time signature, note, symbol, rest, dot, chord, accidental, sharp, flat, natural, key

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*This document corresponds to `mousik` v0.1, dated 2020/07/01.

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1 Introduction

There are already several ways to set music using L^AT_EX. As stated by the `abc` package, “*notably Music_{TE}X and Lilypond*”, as well as `abc per se`. Some of them are too difficult to use for not too complicated music; others do not work at all.

The `mousik` package is perfect for short, one line music fragments. It may be used for longer scores, but there are more adaptable and lighter packages than this one. However, in spite of its compactness, the package is quite complete and versatile, but easy to learn and use.

2 The `mousik` environment

The way to draw a score is by means of the `mousik` environment. For a longer example, see section 5.



```
\begin{mousik}
  \Clef
  \Key{-3}
  \TimeSig{4}{4}
  \Rest{8}
  \Note{16}{7,6}
  \Note{16}{5,2,5}
  \Bar
  \Note{1}{5}
  \Chord
  \Acc{0}{9}
  \Note{1}{9}
  \Bar[|. ]
\end{mousik}
```

2.1 Options

- `indent` indents the first line. Default is 0.
- `long` changes the maximum width of the score. Default is 24.

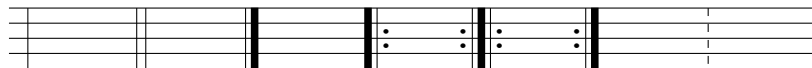
piano scores

space between notes

3 Available commands

3.1 `\Bar`

The `\Bar` command prints a bar on the score. It has an optional argument depending on the type of bar, with default `|`, the single bar (`\Bar = \Bar[|]`).

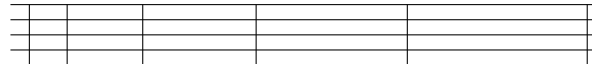


`[|]` `[||]` `[|.]` `[|:]` `[:::]` `[::|]` `[:]` `\Space`

The only way to end a line in the score is with a bar or a space. Any other way may cause visual errors (see subsection 4.1).

3.2 `\Space`

The `\Space` command is not only used when no bar is needed. It may also be used to input *extra* space between symbols. The default space is 2 (which is the default space between symbols in general), but it may be any whole number (see subsection 4.2).



`[-1][0]` `[1]` `[2]` `[3]`

3.3 `\Clef`

Clefs might be used at any moment on the score, although they are automatically printed if an overflow happens (see subsection 4.1). The `\Clef` command has an optional argument which corresponds to its type; the Treble Clef is the default (`\Clef = \Clef[2]`).

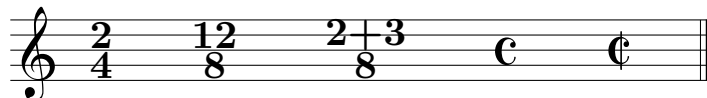


`[1]` `[2]` `[3]` `[4]` `[11]` `[12]` `[13]` `[14]` `[15]` `[0]`

The `\Octave` command prints an eight on a previous clef. It has one argument: if it is an 8, it prints it above the clef; if it is a -8, it prints it below the clef.

3.4 `\TimeSig`

Time signatures might be used at any moment. `\TimeSig{2}{4}` will draw a 2 by 4 time signature. Any number or symbol can be written inside the brackets. Also `\TimeSig{C}{}` will produce a *common time* symbol and `\TimeSig{C|}{}` will produce an *alla breve* symbol.



3.5 Notes and rests

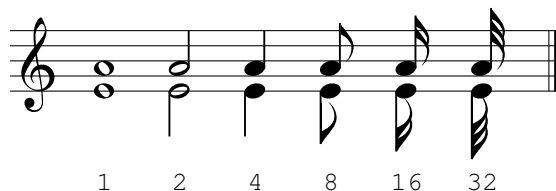
3.5.1 `\Note`

The `\Note` command prints a note on the score. It accepts 2 compulsory arguments, as well as some options:

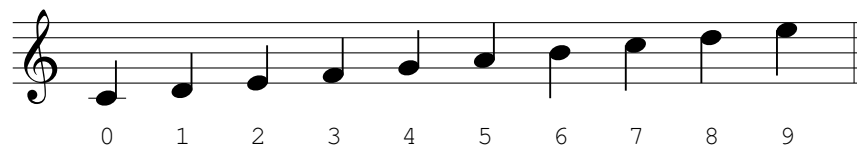
- *beam* chooses the width of the beam. Default is 1 (mm).
- *stem* chooses the height of the stem. Default is 0.3.
- *head* chooses the type of head. Default is “.” (round), but also “x” (cross) and “o” (hollow) are possible.

type of head

The first compulsory argument is the duration of the note.



The second argument is the absolute height. For example, these are the notes from 0 to 9:



The `\Note` command might also print several notes tied together with a beam. Two notes will be tied together with an oblique beam, but three or more will have a straight beam.

monotone triplets sliding beam

The notes go upside down when their height (or maximum height) is bigger than 6. However, for manual control, the `\Swap` command might be used *before* the note.

To write the number of a tuplet, the `\Tuplet` command might be used *after* the note. It has two arguments, an optional one and a compulsory one. The optional one is the written number, which is automatically calculated by default. The compulsory one is the height.



The image shows a musical staff with three groups of notes. The first group has two eighth notes, the second has two eighth notes, and the third has a triplet of eighth notes. Below the staff, the corresponding LaTeX commands are listed: `\Note{16}{5,6}` for the first group, `\Swap\Note{32}{5,6}` for the second group, and `\Note{8}{7,8,9}` followed by `\Tuplet{-5}` for the third group.

bar on tu-
plet, fix
height

Notes with different durations that are tied together have a different syntax. Instead of using 8, 16 and 32 for the duration, each note has to be referenced with a `-`, `a` or `a ≡`, respectively, and additional `|` might be needed for half the beam.



The image shows a musical staff with a sequence of tied notes of different durations: a half note, a quarter note, an eighth note, and a sixteenth note, followed by their respective beams. Below the staff, the LaTeX command is shown: `\Note{|≡,-,|=,=,|=,-,|=,≡|}{0,1,2,3,4,3,2,1}`.

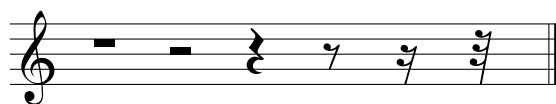
explain bet-
ter

bigger space

To be able to write the `≡` as a symbol on the .tex file, I usually copy and paste it wherever I need it. Nonetheless, an alternative keyboard-friendly input that is equally valid is using `///` instead of `≡`.

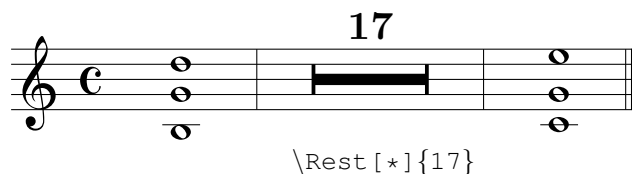
3.5.2 `\Rest`

The `\Rest` command prints a rest. It accepts its duration as an argument: 1, 2, 4, 8, 16 or 32.



The image shows a musical staff with rests of durations 1, 2, 4, 8, 16, and 32. Below the staff, the corresponding LaTeX commands are listed: `1`, `2`, `4`, `8`, `16`, and `32`.

If a multimeasure rest is needed —where the rest lasts an arbitrary amount of bars— an optional argument accepts a `*`.

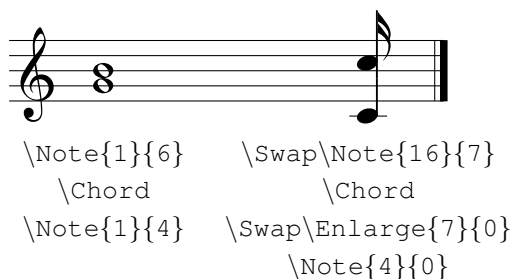


3.5.3 `\Chord`

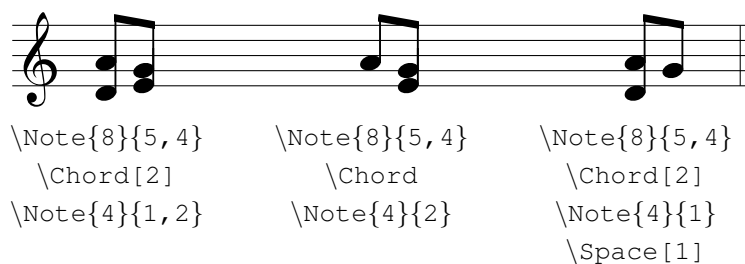
Chords may be obtained by means of the `\Chord` command. It accepts an optional argument: the number of notes to step backwards —mostly for the `\Note` command— with 1 as default.

The best way to use it is to call the `\Note` command with the highest notes; then call the `\Chord` command; then call the `\Note` command with the duration 4 of a quarter note, to print them on top.

If the stem needs to be longer, the user may need the `\Enlarge` command, which accepts as arguments the 2 heights of both notes.



For extreme uses of the command, extra `\Space` commands might be needed.



maybe fix
chord+space

3.6 Keys and accidentals

3.6.1 \Acc

The `Acc` command prints an accidental before a note. It accepts 2 arguments.

The first argument is the type of accidental: a number ranging from -2 to 2. The flat ♭ symbol might also be called with a b and the sharp # symbol might be called with \#.

only # ?

weird width
bb

The second argument is the height of the symbol.

When there are several notes, the `\Accs` command might be useful. It accepts as an argument a list of pairs: type/height. Where there is no accidental, the spot shall be left empty.

$\backslash \text{Accs}\{-2/5, \text{ }, -1/0\}$
 $\backslash \text{Note}\{5, 3, 0\}$

3.6.2 \Key

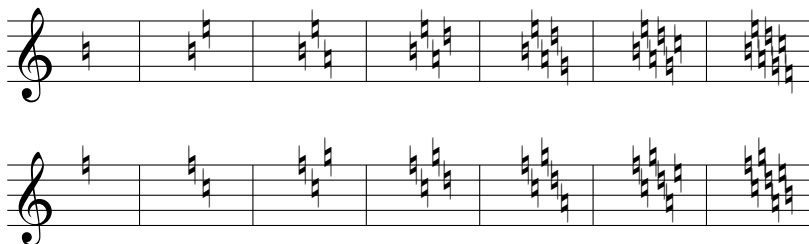
The `\Key` command prints the key at any moment on the score, although they are automatically printed if an overflow happens (see subsection 4.1).

It accepts an argument: a positive whole number from 1 to 7 for sharps $\sharp$, and a negative whole number from -1 to -7 for flats $\flat$.

3.6.3 \NoKey

The `\NoKey` command is the same as the `\Key` command, but it prints naturals  instead of sharps or flats.

Mixed Keys



Be careful: the `\Key` and `\NoKey` commands are affected by the current clef! Nonetheless, other commands such as `\Note` or `\Acc` are not affected and need absolute height values.



```
\begin{mousik}
  \Clef[4]
  \Key{1}
  \TimeSig{C|}{ }

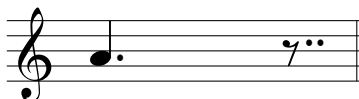
  \Acc{0}{8}
  \Note{1}{8}
  \Bar[|. ]
\end{mousik}
```

3.7 Other symbols

3.7.1 \Dot

The `\Dot` command prints any amount of dots in front of a note or rest. It accepts an optional argument and a compulsory argument.

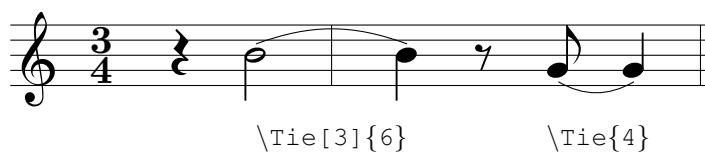
The default number of dots is 1, but it may be any whole number bigger than one. The second, compulsory argument is the height of the dot: `\Dot[2]{7}`.



```
\Dot{5}      \Dot[2]{7}
\Note{4}{5}  \Rest{8}
```

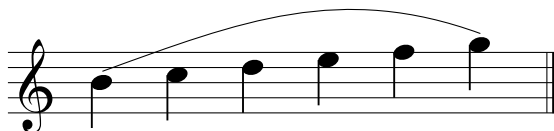
3.7.2 `\Tie`

The `\Tie` command draws a tie between two notes. It has a compulsory argument, the height of both notes, and an optional one: the separation between the two notes. The default value is 2.



3.7.3 The **Slurred** environment

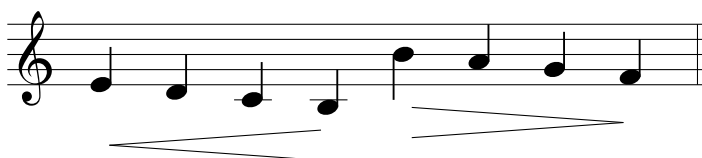
The **Slurred** environment draws a slur between the notes placed inside. It has two arguments: the heights of both beginning and end notes.



Fix angle

3.7.4 The **Crescendo** and **Decrescendo** environments

The **Crescendo** and **Decrescendo** environments draw a crescendo or decrescendo symbol, respectively, between the notes placed inside. It has an optional argument: the height of the symbol. The default value is 0.



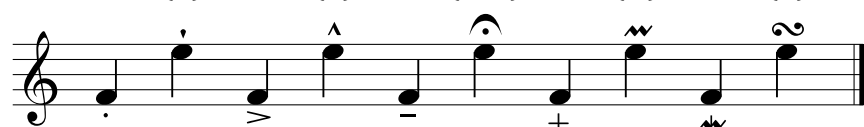
Half line

3.7.5 `\Artic` and `\Symbol`

The `\Artic` command prints an articulation symbol on a successive note. It accepts two arguments. The first one is the type of articulation.

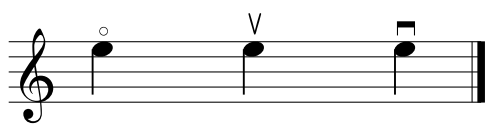
Articsss

Staccatissimo	Marcato	Fermata	Mordent	Turn
{'}	{^}	{(.)}	{~}	{?}



{.}	{>}	{-}	{+}	{~ }
Staccato	Accent	Tenuto	Trill	Lower mordent

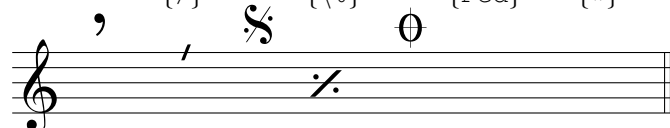
Harmonic	Up bow	Down bow
{o}	{v}	{n}



The second one is the height of the symbol, which might or might not be the height of the corresponding note.

The `\Symbol` is similar to the `\Artic` command, but it only accepts the type of symbol and not the height. These are symbols that have a fixed position on the score.

Caesura	Simile	Pedal	Lift Pedal
{/}	{\%	{Ped}	{*}



{,}	{s\%	{o+}	Ped.	*
Breath	Segno	Coda		

3.8 `\Text` and `\Dynamics`

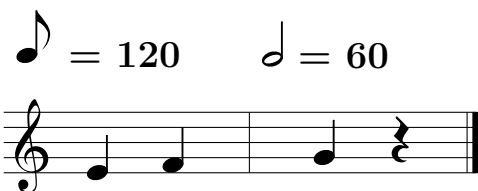
The `\Text` command allows the printing of any text string. It has a compulsory argument, but also an optional one where the user may choose the height of the string: `\Text[2]{\it rit.}`.

To write anything in the usual font of dynamics, the `\Dynamics` command will work the same: `\Dynamics[-0.5]{pp}`.



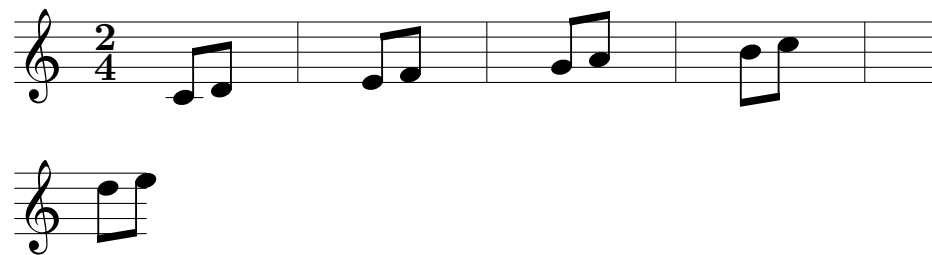
3.9 \Tempo

The `\Tempo` command prints a metronome tempo indicator. It has two arguments: the duration of the note and the numerical tempo which will be printed.



4 Tips and observations

4.1 Overflow of lines



explain over-
flow

4.2 How do spaces work?

Spaces serve as the manual progression of an invisible cursor during the printing. A negative

explain
spaces

4.3 What if it becomes too cumbersome: your own commands

If a concrete rhythm is constantly used on a music piece.

explain

4.4 On your own: working with the `tikz` package

explain tikz

5 A very long and complete example

choose example

